

Symmetric Cipher Model

Chapter-2: Classical Encryption Techniques

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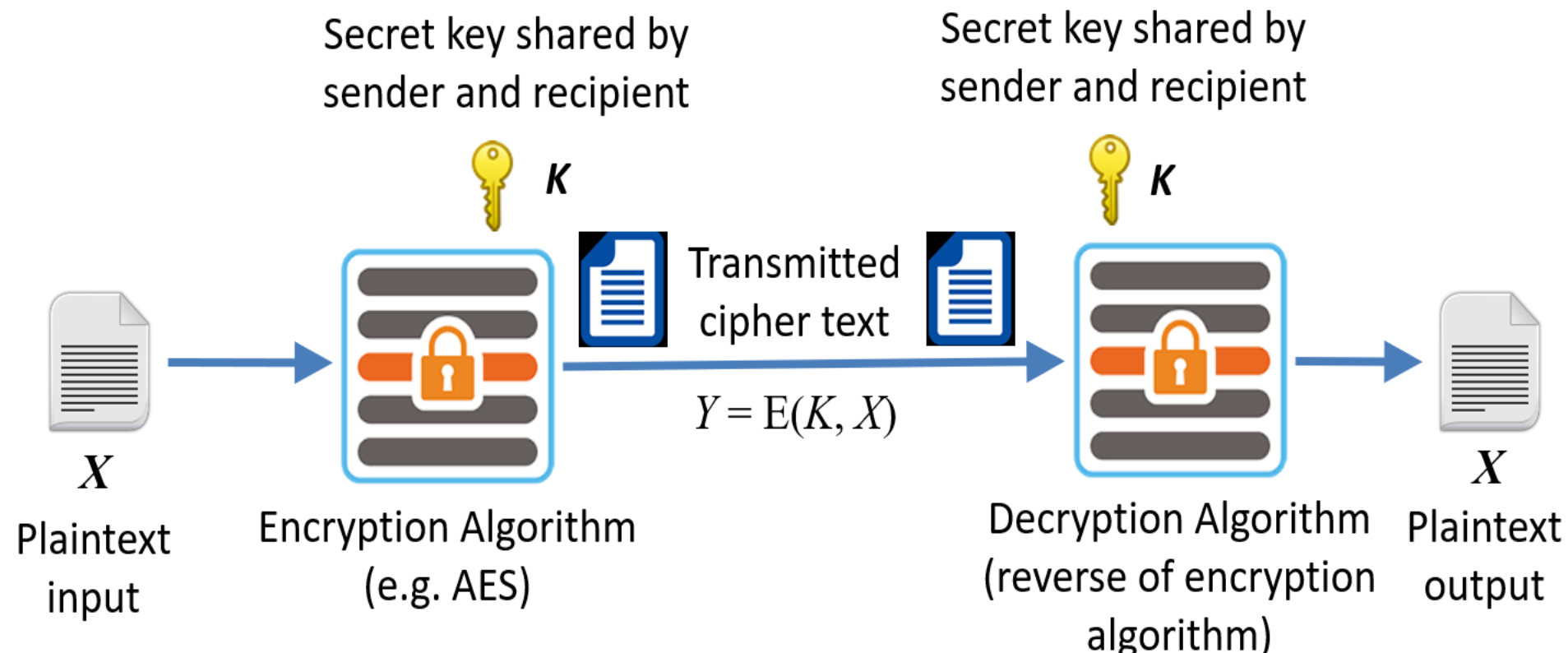
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Symmetric Cipher Model



Symmetric Cipher Model

1. An original message is known as the **plaintext**, while the coded message is called the ciphertext.
2. The process of converting from plaintext to ciphertext is known as **enciphering or encryption**; restoring the plaintext from the ciphertext is deciphering or decryption.
3. Plaintext is the original intelligible message or data that is fed into the algorithm as input.
4. Encryption algorithm performs various substitutions and transformations on the plaintext.

Symmetric Cipher Model

5. The **secret key** is also input to the encryption algorithm. The key is a value independent of the plaintext and of the algorithm.
6. **Decryption algorithm** is essentially the encryption algorithm run in reverse. It takes the ciphertext and the secret key and produces the original plaintext.
7. Ciphertext is the scrambled message produced as output. It depends on the plaintext and the secret key.
8. The ciphertext is an apparently random stream of data and, as it stands, is unintelligible.

Impact of Security Breach :

- **Plaintext:** original message
- **Ciphertext:** encrypted or coded message
- **Encryption:** convert from plaintext to ciphertext (enciphering)
- **Decryption:** restore the plaintext from ciphertext (deciphering)
- **Key:** information used in cipher known only to sender/receiver
- **Cipher:** a particular algorithm (cryptographic system)
- **Cryptography:** study of algorithms used for encryption
- **Cryptanalysis:** study of techniques used for decryption without knowledge of plaintext
- **Cryptology:** areas of cryptography and cryptanalysis

Cryptanalysis

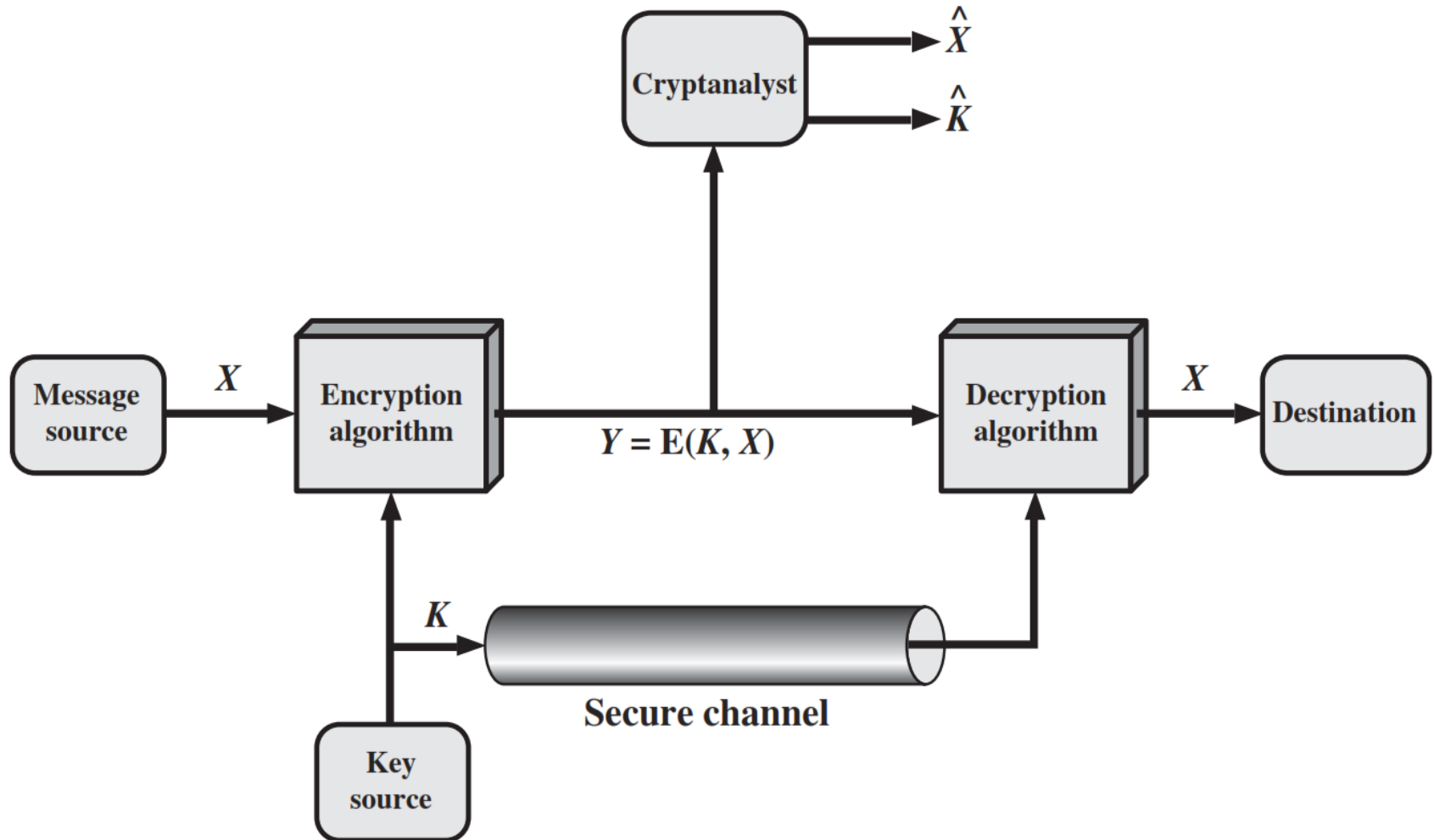
1. Cryptography(Secret Writing) is the process of protecting information by transforming it into a secure (unreadable) format.



2. Cryptanalysis is the decryption and analysis of encrypted text. Cryptanalysis uses mathematical formulas to search algorithm vulnerabilities and break into cryptography.



Cryptanalysis



Cryptanalysis

An opponent, observing Y but not having access to K or X , may attempt to recover X or K or both X and K .

If the opponent is interested in only this particular message, then he will focus to recover X by generating a plaintext estimate $\hat{X}$.

Often, however, the opponent is interested in being able to read future messages as well, in which case an attempt is made to recover K by generating an estimate $\hat{K}$.

Requirement and Assumption

Requirements for secure use of symmetric encryption:

- **Strong encryption algorithm:** Given the algorithm and cipher text, an attacker cannot obtain key or plaintext.
- **Shared secret keys:** sender and receiver both have shared a secret key; no-one else knows the key(keep it secret).

Assumptions:

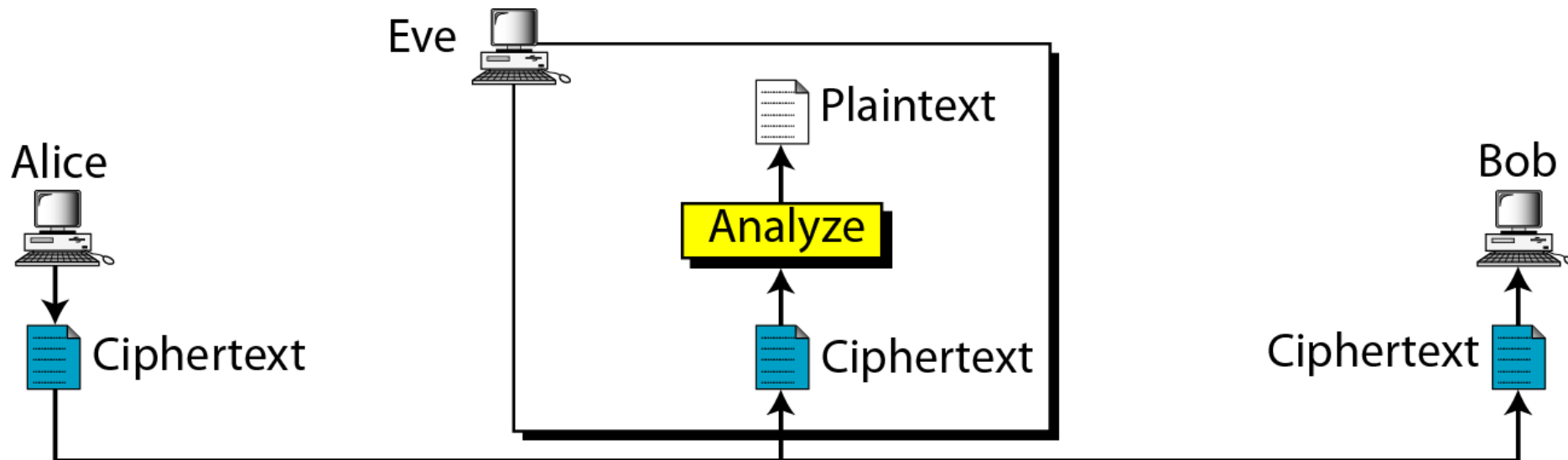
- Cipher is known
- Secure channel to distribute keys

Cryptanalysis and Brute force Attack

- Objective of attacker: recover key (not just message)
- Approaches of attacker:
- Cryptanalysis: This type of attack exploits the characteristics of the algorithm to attempt to derive a specific plaintext or to derive the key being used.
- Brute-force attack: The attacker tries every possible key on a piece of ciphertext until an intelligible translation into plaintext is obtained.
- On average, half of all possible keys must be tried to achieve success.

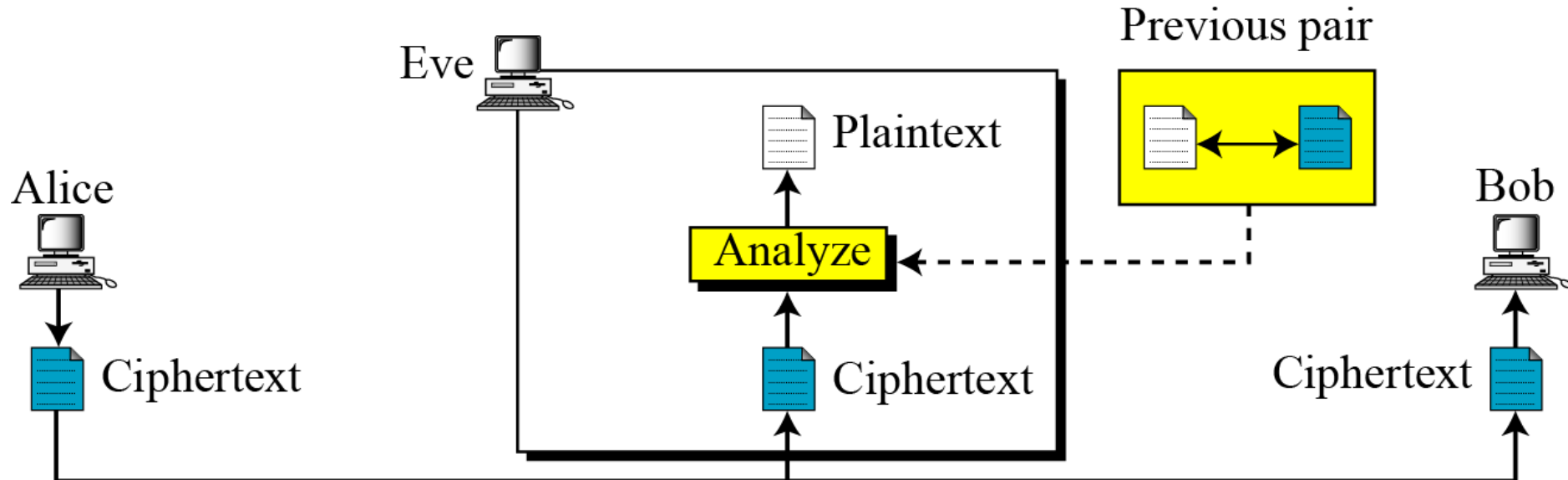
Cryptanalysis Attacks

Type of Attack	Known to cryptanalyst
Ciphertext Only	Encryption algorithm, Ciphertext



Cryptanalysis Attacks

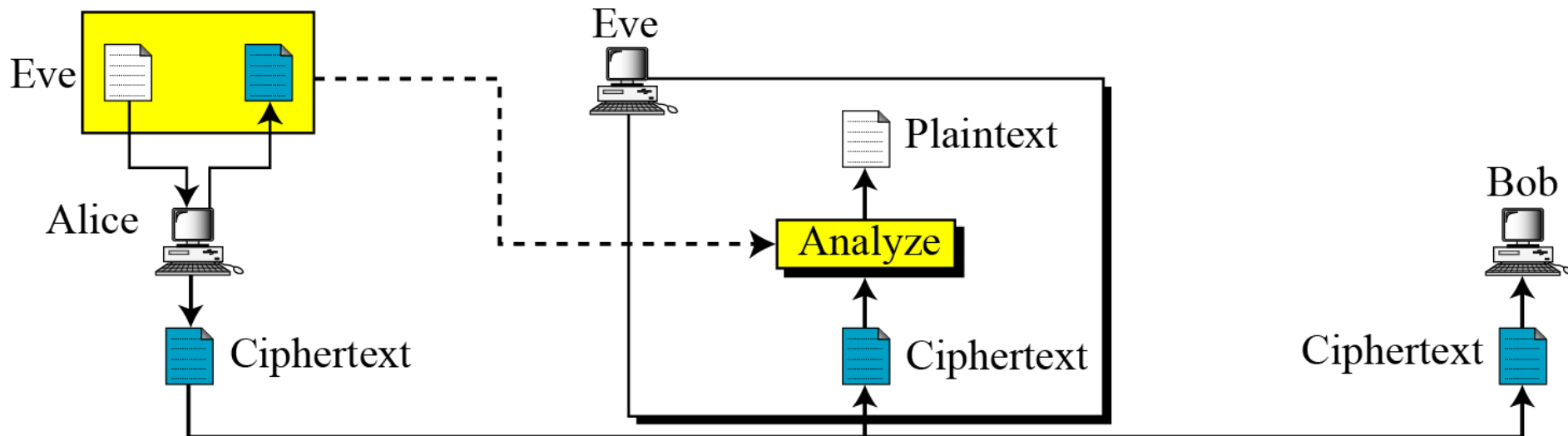
Type of Attack	Known to cryptanalyst
Known Plaintext	Encryption algorithm, Ciphertext, One or more plaintext-cipher text pairs formed with the secret key



Cryptanalysis Attacks

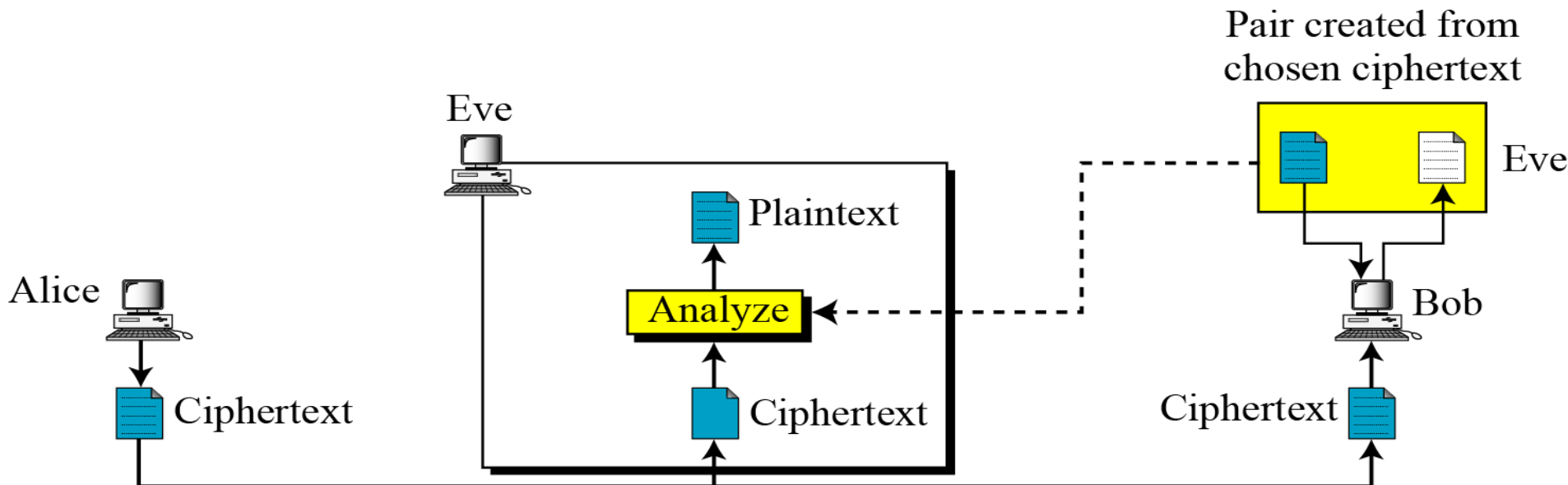
Type of Attack	Known to cryptanalyst
Chosen Plaintext	Encryption algorithm, Ciphertext, Plaintext message chosen by cryptanalyst

Pair created from
chosen plaintext



Cryptanalysis Attacks

Type of Attack	Known to cryptanalyst
Chosen Ciphertext	Encryption algorithm, Ciphertext, Ciphertext chosen by cryptanalyst, with its corresponding decrypted plaintext generated with the secret key

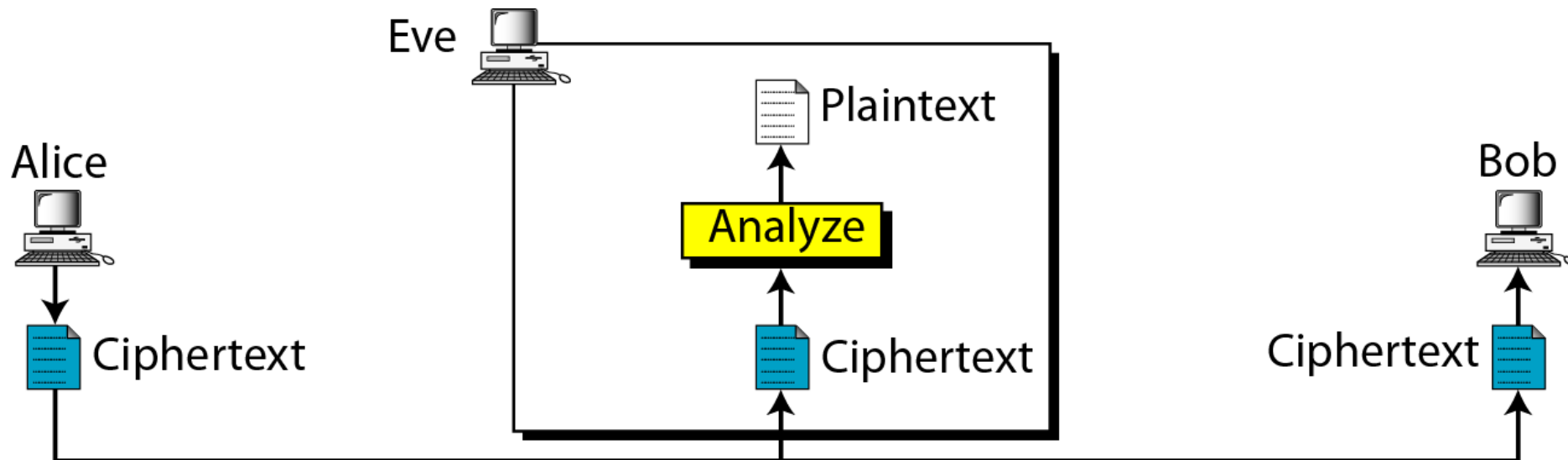


Cryptanalysis Attacks

Type of Attack	Known to cryptanalyst
Chosen text	Encryption algorithm, Ciphertext, Plaintext chosen by cryptanalyst, with its corresponding ciphertext generated with the secret key , Ciphertext chosen by cryptanalyst, with its corresponding decrypted plaintext generated with the secret key

Cryptanalysis Attacks

Type of Attack	Known to cryptanalyst
Ciphertext Only	Encryption algorithm, Ciphertext



Substitution Technique

A substitution technique is one in which the letters of plaintext are replaced by other letters or by numbers or symbols.

If plaintext viewed as sequence of bits, replace plaintext bit patterns with ciphertext bit patterns.

1. Caesar Cipher
2. Monoalphabetic Cipher
3. Playfair Cipher
4. Hill Cipher
5. Polyalphabetic Ciphers
6. One-Time Pad

Ceaser Cipher

The Caesar Cipher involves replacing each letter of the alphabet with the letter standing three places further down the alphabet.

For encryption algorithm is:

$$C = E(3, P) = (P + 3) \bmod 26$$

For decryption algorithm is

$$P = D(3, C) = (C - 3) \bmod 26$$

Ceaser Cipher

Let us assign a numerical equivalent to each letter

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

$$C = E(3, P) = (P + 3) \bmod 26$$

Plain: a b c d e f g h i j k l m n o p q r s t u v w x y z

Cipher: d e f g h i j k l m n o p q r s t u v w x y z a b c

Ceaser Cipher

Example:

Plaintext: THE QUICK BROWN FOX

Ciphertext: WKH TXLFN EURZQ IRA

Generalized Ceaser Cipher

Allow shift by k positions.

Encryption : $C = E(K, P) = (P + K) \bmod 26$

Decryption : $P = D(K, C) = (C - K) \bmod 26$

Modulo for negative number is $= N - (B \% N)$

Example :

$$-11 \bmod 26 = 15$$

$$26 - (11 \% 26) = 15$$

Ceaser Cipher

1. Plaintext: networksecurity
Key: 7
2. Cipher: exxegoexsrgi
Key: 4
3. Cipher: kyzj dvjjrxv zj vetipgkvu
Key: 17
4. Plain: information security
Key: 16

Ceaser Cipher

1. **Plaintext:**
networksecurity

Key: 7

Cipher:
uladvyrzljbypaf

2. **Cipher:** exxegoexsrgi

Key: 4

Plaintext:
Attackatonce

3. **Cipher:** kyzj dvjjrxv zj vetipgkvu
Key: 17

Plaintext: this message is
encrypted

4. **Plaintext:** information security

Key: 16

Ciphertext: tyqzcxletzy dpnfctej

Brute force attack on Ceaser Cipher

The encryption and decryption algorithms are known.

There are only 25 keys to try, e.g. $k=1$, $k=2$, ...

The language of the plaintext is known and easily recognizable.

Ceaser Cipher

Ciphertext: ZNK WAOIQ HXUCT LUD

Key	Transformed text
1	YMJ VZNHP GWTBS KTC
2	XLI UYMGO FVSAR JSB
3	WKH TXLFN EURZQ IRA
4	VJG SWKEM DTQYP HQZ
5	UIF RVJDL CSPXOGPY
6	THE QUICK BROWN FOX
7	SGD PTHBJ AQNVM ENW
8	RFC OSGAI ZPMUL DMV
9	QEB NRFZH YOLTK CLU
10	PDA MQEYG XNKSJ BKT
11	OCZ LPDXF WMJRI AJS
12	NBY KOCWE VLIQH ZIR
13	MAX JNBVD UKHPG YHQ

Key	Transformed text
14	LZW IMAUC TJGOF XGP
15	KYV HLZTB SIFNE WFO
16	JXU GKYSR RHEMD VEN
17	IWT FJXRZ QGDLC UDM
18	HVS EIWQY PFCKB TCL
19	GUR DHVPX OEBJA SBK
20	FTQ CGUOW NDAIZ RAJ
21	ESP BFTNV MCZHY QZI
22	DRO AESMU LBYGX PYH
23	CQN ZDRLT KAXFW OXG
24	BPM YCQKS JZWEV NWF
25	AOL XBPJR IYVDU MVE

Monoalphabetic Cipher

It is an improvement to the Caesar Cipher.

Instead of shifting the alphabets by some number, this scheme uses some permutation of the letters in alphabet.

Use a single alphabet for both plaintext and cipher text.

Example:

Cipher: kxlvzofemrj

Plaintext: ?

Try Brute force attack :

With 26 letters in alphabet, the possible permutations are 26! Keys (>4x10²⁶)

Attack on Monoalphabetic Cipher

- The relative frequencies of the letters in the ciphertext (in %) are

P 13.33	H 5.83	F 3.33	B 1.67	C 0.00
Z 11.67	D 5.00	W 3.33	G 1.67	K 0.00
S 8.33	E 5.00	Q 2.50	Y 1.67	L 0.00
U 8.33	V 4.17	T 2.50	I 0.83	N 0.00
O 7.50	X 4.17	A 1.67	J 0.83	R 0.00
M 6.67				

Ciphertext:

uzqsovuohxmopvgpozpevsgzwszopfpesxudbmetsxaizvuephzhmdzshz
owsfpappdtsvpquzwxymxuzuhxsxepyepopdzszufpombzwpfupzhmdjudt
mohmq

Attack on Monoalphabetic Cipher

In our ciphertext, the most common digram is ZW, which appears three times. So equate Z with t, W with h and P with e.

Now notice that the sequence ZWP appears in the ciphertext, and we can translate that sequence as “the.”

If the cryptanalyst knows the nature of the plaintext, then the analyst can exploit the regularities of the language.

The relative frequency of the letters can be determined and compared to a standard frequency distribution for English.

If the message were long enough, this technique alone might be sufficient, but because this is a relatively short message, we cannot expect an exact match.

Hill Cipher

Hill cipher is based on linear algebra

Each letter is represented by numbers from 0 to 25 and calculations are done using modulo 26.

Encryption and decryption can be given by the following formula:

Encryption: $C = PK \bmod 26$

Decryption: $P = CK^{-1} \bmod 26$

$$(c_1 \ c_2 \ c_3) = (p_1 \ p_2 \ p_3) \begin{pmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{pmatrix} \bmod 26$$

Hill Cipher

To encrypt a message using the Hill Cipher we must first turn our keyword and plaintext into a matrix (a 2 x 2 matrix or a 3 x 3 matrix, etc).

Example: Key = "HILL" Plaintext = "EVAM"

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

$$\text{Key Matrix} \begin{pmatrix} H & I \\ L & L \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$$

$$\text{Plaintext} \begin{pmatrix} E \\ X \end{pmatrix} \begin{pmatrix} A \\ M \end{pmatrix} = \begin{pmatrix} 4 \\ 23 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix}$$

Hill Cipher

$$\text{Key Matrix } \begin{pmatrix} H & I \\ L & L \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$$

$$\text{Plaintext } \begin{pmatrix} E \\ X \end{pmatrix} \begin{pmatrix} A \\ M \end{pmatrix} = \begin{pmatrix} 4 \\ 23 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix}$$

$$C = PK \text{ mod } 26$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 4 \\ 23 \end{pmatrix}$$

$$7 \times 4 + 8 \times 23 = 212$$

$$11 \times 4 + 11 \times 23 = 297$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 4 \\ 23 \end{pmatrix} = \begin{pmatrix} 212 \\ 297 \end{pmatrix}$$

$$\begin{pmatrix} 212 \\ 297 \end{pmatrix} = \begin{pmatrix} 4 \\ 11 \end{pmatrix} \text{ mod } 26 = \begin{pmatrix} E \\ L \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix}$$

$$7 \times 0 + 8 \times 12 = 96$$

$$11 \times 0 + 11 \times 12 = 132$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix} = \begin{pmatrix} 96 \\ 132 \end{pmatrix}$$

$$\begin{pmatrix} 96 \\ 132 \end{pmatrix} = \begin{pmatrix} 18 \\ 2 \end{pmatrix} \text{ mod } 26 = \begin{pmatrix} S \\ C \end{pmatrix}$$

Ciphertext = "ELSC"

Hill Cipher

$$P = CK^{-1} \text{ mod } 26$$

Step 1: Find Inverse of key matrix

Step 2: Multiply the Multiplicative Inverse of the Determinant by the Adjoin Matrix

Step 3: Multiply inverse key matrix with ciphertext matrix to obtain plaintext matrix

Hill Cipher

Step 1: Inverse of key matrix

2 X 2 inverse of matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - cb} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

3 X 3 inverse of matrix

$$A^{-1} = \frac{1}{\text{determinant}(A)} \cdot \text{adjoin}(A)$$

$$\text{Inverse Key Matrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}^{-1} = \frac{1}{77 - 88} \begin{pmatrix} 11 & -8 \\ -11 & 7 \end{pmatrix}$$

$$= \frac{1}{-11} \begin{pmatrix} 11 & -8 \\ -11 & 7 \end{pmatrix}$$

$$= \frac{1}{15} \begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix} \text{ mod } 26$$

- $-11 \text{ mod } 26 = 15$
- Because, modulo for negative number is $= N - (B\%N)$
 $= 26 - (11\%26)$

Hill Cipher

Step 2: Modular (Multiplicative) inverse

The inverse of a number A is $1/A$ since $A * 1/A = 1$

e.g. the inverse of 5 is $1/5$

In modular arithmetic we do not have a division operation.

The modular inverse of A (mod C) is A^{-1}

$$(A * A^{-1}) \equiv 1 \pmod{C}$$

Example:

The modular inverse of A mod C is the B value that makes

$$A * A^{-1} \pmod{C} = 1$$

$$A = 3, C = 11$$

Since $(3 * 4) \pmod{11} = 1$, 4 is modulo inverse of 3

$$A = 10, C = 17, A^{-1} = 12$$

Hill Cipher

Step 2: Modular (Multiplicative) inverse

Determinants' multiplicative inverse Modulo 26												
Determinant	1	3	5	7	9	11	15	17	19	21	23	25
Inverse Modulo 26	1	9	21	15	3	19	7	23	11	5	17	25

$$= \frac{1}{15} \begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix} \text{ mod } 26$$

- Multiplicative inverse of $\frac{1}{15}$ is 7

Hill Cipher

Step 3: Multiply with adjoint of matrix

$$= 7 \begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix} = \begin{pmatrix} 77 & 126 \\ 105 & 49 \end{pmatrix} = \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \text{ mod } 26$$

$$= \text{thus, if } K = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \text{ then } K^{-1} = \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix}$$

$$\begin{aligned} X \% Y &= X - (X/Y) * Y \\ 77 \% 26 &= 77 - (77/26) * 26 \\ &= 77 - (2) * 26 \\ &= 77 - 52 \\ &= 25 \end{aligned}$$

Hill Cipher

$$\text{Inverse Key Matrix} = \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix}$$

$$\text{Ciphertext} \begin{pmatrix} E \\ L \end{pmatrix} \begin{pmatrix} S \\ C \end{pmatrix} = \begin{pmatrix} 4 \\ 11 \end{pmatrix} \begin{pmatrix} 18 \\ 2 \end{pmatrix}$$

$$P = CK^{-1} \text{ mod } 26$$

$$\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 4 \\ 11 \end{pmatrix}$$

$$25 \times 4 + 22 \times 11 = 342$$

$$1 \times 4 + 23 \times 11 = 257$$

$$\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 4 \\ 11 \end{pmatrix} = \begin{pmatrix} 342 \\ 257 \end{pmatrix}$$

$$\begin{pmatrix} 342 \\ 257 \end{pmatrix} = \begin{pmatrix} 4 \\ 23 \end{pmatrix} \text{ mod } 26 = \begin{pmatrix} E \\ X \end{pmatrix}$$

$$\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 18 \\ 2 \end{pmatrix}$$

$$25 \times 18 + 22 \times 2 = 494$$

$$1 \times 18 + 23 \times 2 = 64$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix} = \begin{pmatrix} 494 \\ 64 \end{pmatrix}$$

$$\begin{pmatrix} 494 \\ 64 \end{pmatrix} = \begin{pmatrix} 0 \\ 12 \end{pmatrix} \text{ mod } 26 = \begin{pmatrix} A \\ M \end{pmatrix}$$

Plaintext = "EXAM"

Hill Cipher

Key: Hill

Plaintext: short example

Ciphertext: APADJ TFTWLFJ

Key: ACBA

Plaintext: DR GREER ROCKS (A=1, B=2, ...)

Ciphertext: FZIFTOTBXGPO

Key: DACB

Ciphertext: SAKNOXAOJ (A=1, B=2, ...)

Plaintext: WELOVEMATH

Playfair Cipher

The Playfair algorithm is based on a 5×5 matrix (key) of letters. The matrix is constructed by filling in the letters of the keyword (minus duplicates) from left to right and from top to bottom, and then filling in the remainder of the matrix with the remaining letters in alphabetic order. The letters I and J count as one letter

Example:

Keyword= OCCURRENCE

Plaintext= TALL TREES

O	C	U	R	E
N	A	B	D	F
G	H	I/J	K	L
M	P	Q	S	T
V	W	X	Y	Z

Playfair Cipher

- Operate on pair of letters (digram) at a time.
- Special: if digram with same letters appears, separate by special letter (e.g. x)

Plaintext= TALL TREES

Plaintext= TA LX LT RE ES

- If there is an odd number of letters, then add uncommon letter to complete digram, a X/Z may be added to the last letter.
- Plaintext= NETWORK
- Plaintext= NE TW OR KX

Playfair Cipher

Map each pair in key matrix

Plaintext: TA LX LT RE ES

Ciphertext: PF IZ TZ EO RT

O	C	U	R	E
N	A	B	D	F
G	H	I/J	K	L
M	P	Q	S	T
V	W	X	Y	Z

If the letters appear on the **same row**, replace them with the letters to their immediate right respectively, wrapping around to the left side of the row if necessary.

For example, using the table above, the letter pair RE would be encoded as EO.

Playfair Cipher

If the letters appear on the **same column**, replace them with the letters immediately below, wrapping around to the top if necessary. For example, using the table above, the letter pair LT would be encoded as TZ.

If the letters are on **different rows and columns**, replace them with the letters on other corner of the same row.

The order is important - the first letter of the pair should be replaced first.

For example, using the table above, the letter pair TA would be encoded as PF.

Playfair Cipher – is it breakable?

Better than monoalphabetic: relative frequency of digrams much less than of individual letters.

But relatively easy (digrams, trigrams, expected words)

Playfair Cipher

Key= "engineering "

Key= "keyword "

Key= "moonmission "

Plaintext= "test this process "

Plaintext= "come to the window"

Plaintext= "greet "

E N G I R	Encrypted Message: pi tu pm gt ue lf gp xg
A B C D F	
H K L M O	
P Q S T U	
V W X Y Z	

K E Y W O	Encrypted Message: Lc nk zk vf yo gq ce bw
R D A B C	
F G H I L	
M N P Q S	
T U V X Z	

M O N I S	Encrypted Message: hq cz du
A B C D E	
F G H K L	
P Q R T U	
V W X Y Z	

Playfair Cipher

4. Key: EXAMPLE

Ciphertext: UA ARBED EXAPO PR QNX AXANR

E	X	A	M	P
L	B	C	D	F
G	H	I/J	K	N
O	Q	R	S	T
U	V	W	Y	Z

Pair: UA AR BE DE XA PO PR QN XA XA NR

Plaintext: we wi lx lm ex et at th ex ex it

Plaintext: we wilxl mexet at thex exit

Plaintext: we will meet at the exit

Polyalphabetic Cipher

Monoalphabetic cipher encoded using only one fixed alphabet,

Polyalphabetic cipher is a substitution cipher in which the cipher alphabet for the plain alphabet may be different at different places during the encryption process.

1. Vigenere cipher
2. Vernam cipher

Polyalphabetic Cipher

Plaintext

PT = **HELLO**

KEY = **GMGMG**

CT = **NQRXU**

Key

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Polyalphabetic Cipher - Vigenere cipher

Keyword : **DECEPTIVE**

Key : **DECEPTIVEDECEPTIVEDECEPTIVE**

Plaintext : **WEAREDISCOVEREDSAVEYOURSELF**

Ciphertext : **ZICVTWQNGRZGVTWAVZHCQYGLMGJ**

Key must be as long as plaintext else repeat a keyword

C

$$\begin{aligned} \bar{P} &= (P_1 + K_1, P_2 + K_2, \dots, P_m + K_m) \bmod 26 \\ &= (C_1 - K_1, C_2 - K_2, \dots, C_m - K_m) \bmod 26 \end{aligned}$$

An analyst looking at only the ciphertext would detect the repeated sequences VTW at a displacement of 9 and make the assumption that the keyword is either three or nine letters in length.

Keyword : **DECEPTIVE**

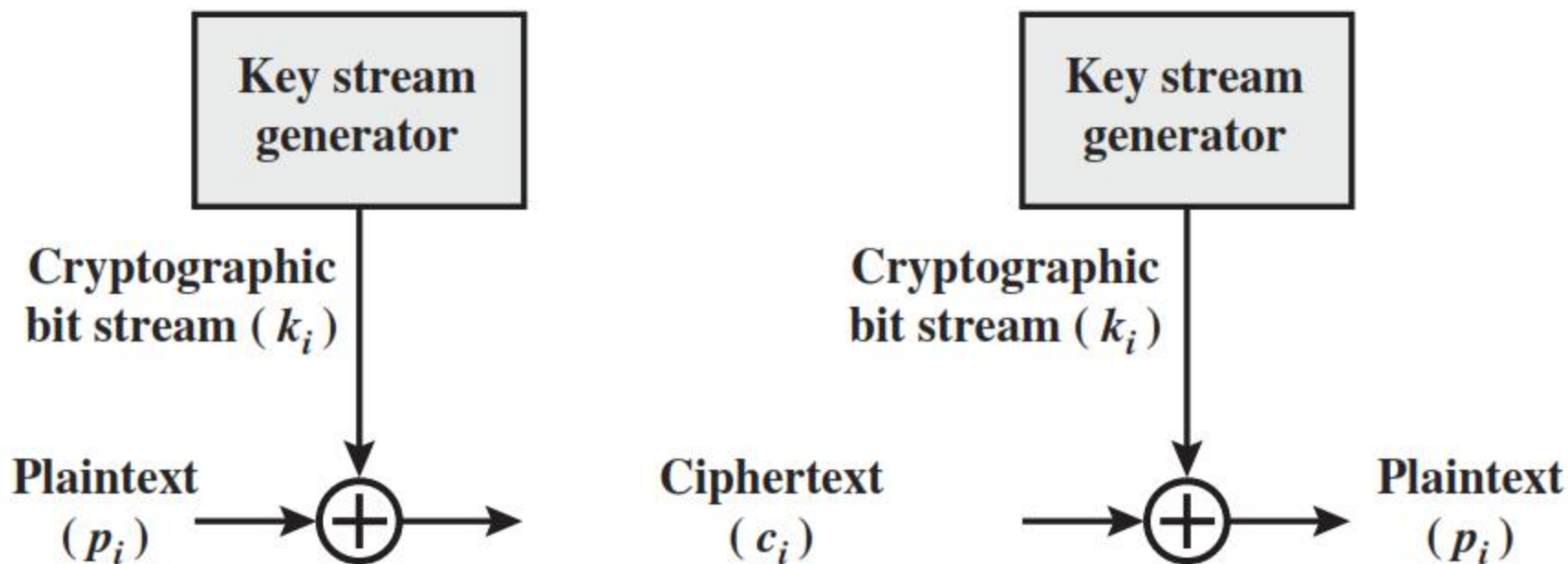
Key : **DECEPTIVEWEAREDISCOVEREDSAV**

Plaintext : **WEAREDISCOVEREDSAVEYOURSELF**

This system is referred as an auto key system

Polyalphabetic Cipher – Vernam cipher

The ciphertext is generated by applying the logical XOR operation to the individual bits of plaintext and the key stream.



OTP – One time pad

Similar to Vigenere, but use random key as long as plaintext.

Only known scheme that is unbreakable (unconditional security)

Ciphertext has no statistical relationship with plaintext.

Given two potential plaintext messages, attacker cannot identify the correct message.

Two practical limitations:

Difficult to provide large number of random keys

Distributing unique long random keys is difficult

OTP – One time pad

Attacker knows the ciphertext:

ANKYODKYUREPFJBYOJDSPLREYIUNOFDOIUERFPLUYTS

Attacker tries all possible keys.

Two examples:

key1: pxlmvmsydofuyrvzwctnlebnecvgdupahfzzlmnyih

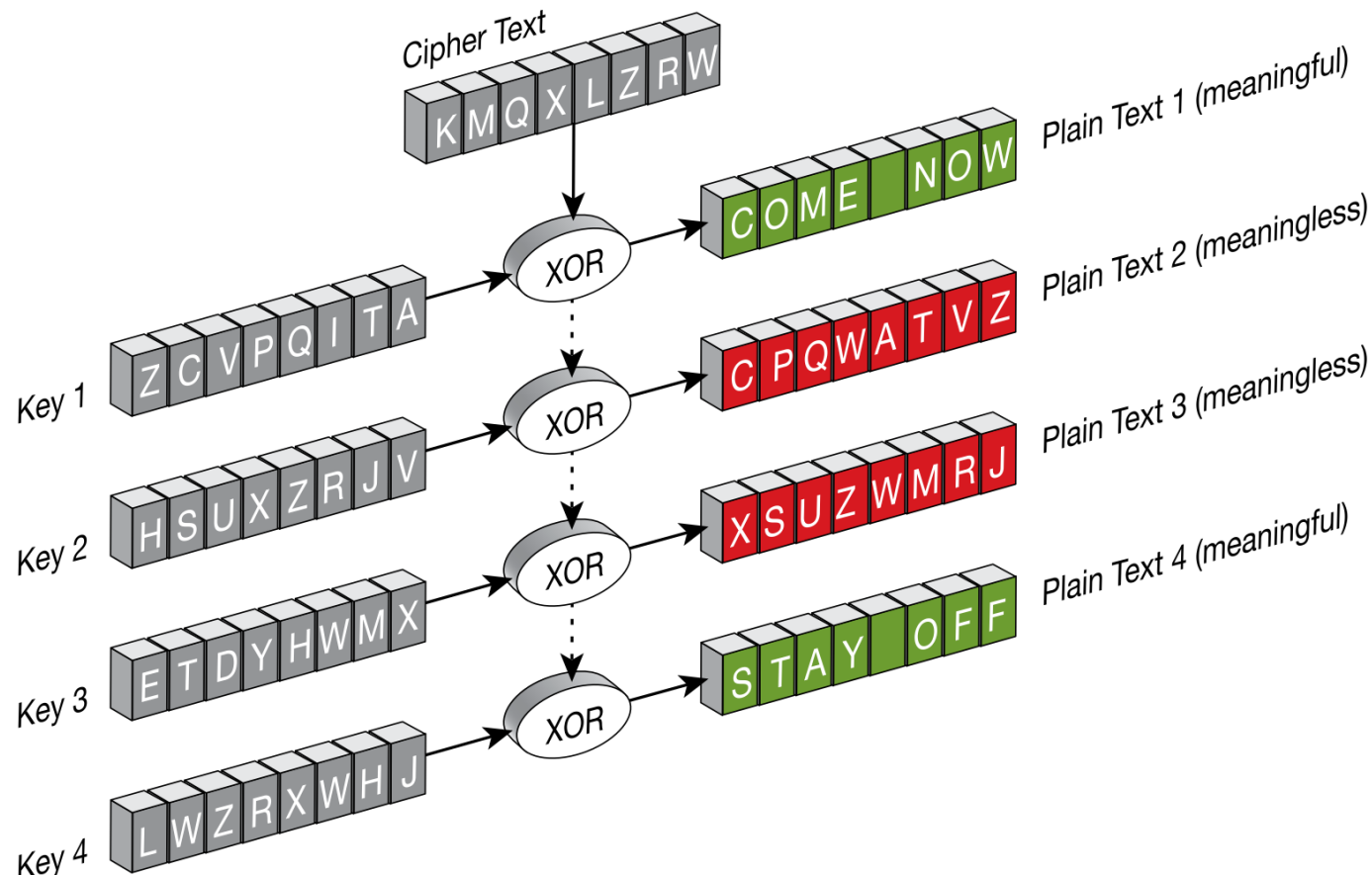
Plaintext1: mr mustard with the candlestick in the hall

key2: mfugpmiydgaxgoufhklmhsqdqogtewbqfgyovuhwt

Plaintext2: miss scarlet with the knife in the library

There are many other legible plaintexts obtained with other keys. No way for attacker to know the correct plaintext

OTP – One time pad

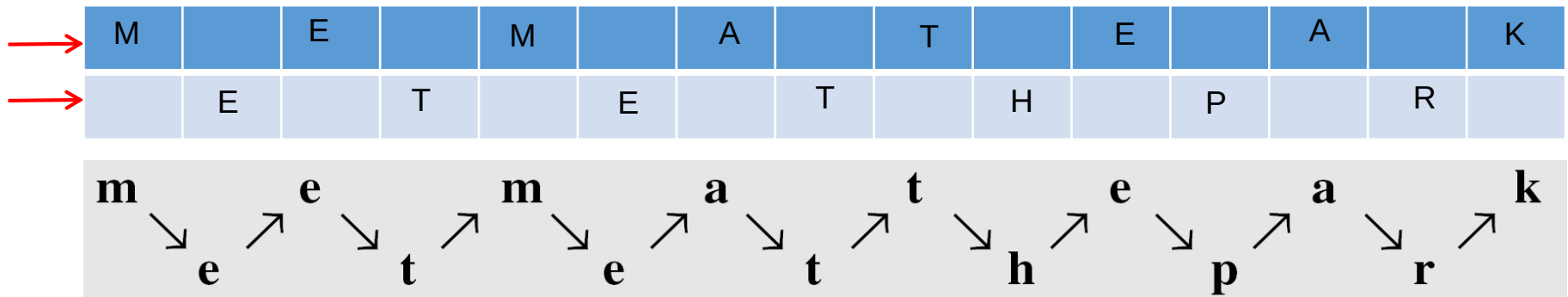


Transposition Technique – Rail Fence Technique

A transposition cipher does not substitute one symbol for another, instead it changes the location of the symbols.

The simplest such cipher is the rail fence technique, in which the plaintext is written down as a sequence of diagonals and then read off as a sequence of rows.

For example, to send the message “Meet me at the park” to Bob, Alice writes



She then creates the ciphertext: MEMATEAK ETETHPR

Transposition Technique – Rail Fence Technique

Easy to break: letter frequency analysis to determine depth.

Example:

Plaintext: internettechnology

Depth: 3

Ciphertext: NGNENTEHOOYTECL

I				R				T				N				G		
	N		E		N		T		E		H		O		O		Y	
		T				E				C				L				

Transposition Technique – Row / Column Transposition

Plaintext letters written in rows.

Ciphertext obtained by reading column-by-column, but re-arranged.

Key determines order of columns to read.

Key:	4 3 1 2 5 6 7
Plaintext:	A T T A C K P
	O S T P O N E
	D U N T I L T
	W O A M X Y Z

Ciphertext: TTNA APTM TSUO AODW COIX KNLY PETZ

Easy to break using letter frequency (try different column orders)

Transposition Technique – Row / Column Transposition

Transposition ciphers can be made stronger by using multiple stages of transposition

Plaintext: securityandcryptography

Key: 315624

Ciphertext:

Transpose again using same key:

Ciphertext:

Steganography

Steganography is the practice of hiding a secret message within a normal-looking file (like an image, audio, or video) so that no one suspects anything is hidden.

It's different from encryption — encryption makes data unreadable; steganography makes data invisible.

Common uses: secure communication, digital watermarking, copyright protection.

Steganography

A cover medium (e.g., a picture) is selected.

The secret message is hidden inside the cover using small changes, often imperceptible to humans.

Example method: **Least Significant Bit (LSB)** — hide information by tweaking the smallest bits of image pixels.

Original pixel: RGB(101, 150, 200)

After hiding data: RGB(101, 151, 200) → Barely noticeable!

Steganography

Example: Hiding "Hi" inside an image

"Hi" in binary: H -> 01001000, i -> 01101001

Modify 16 pixels' least significant bits to match these bits.

The image looks **exactly the same** to the naked eye!

Tools you can use:

- Online steganography apps
- Python libraries like Stegano or Pillow

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